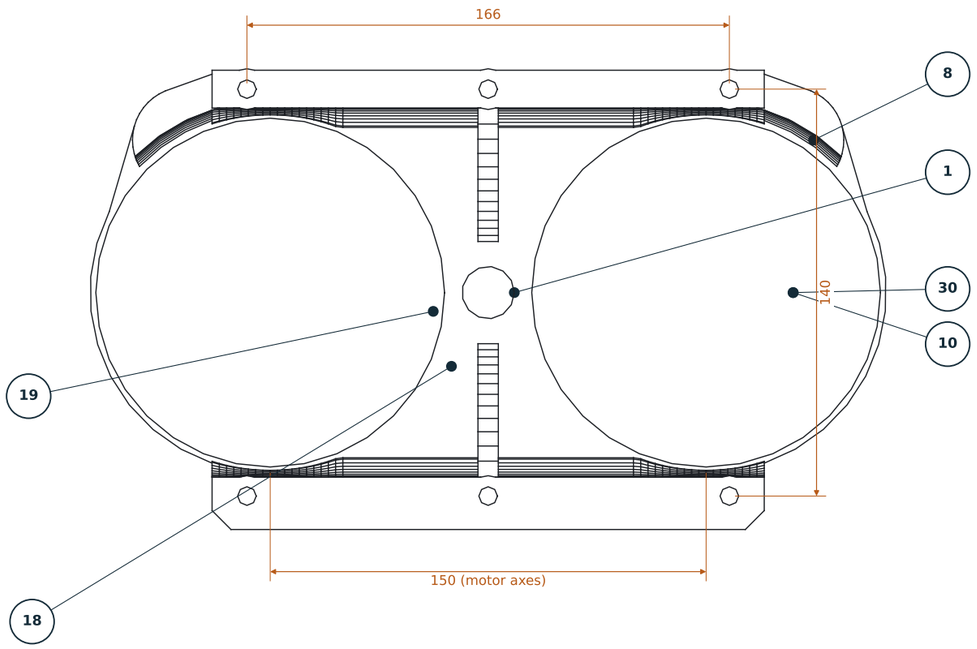
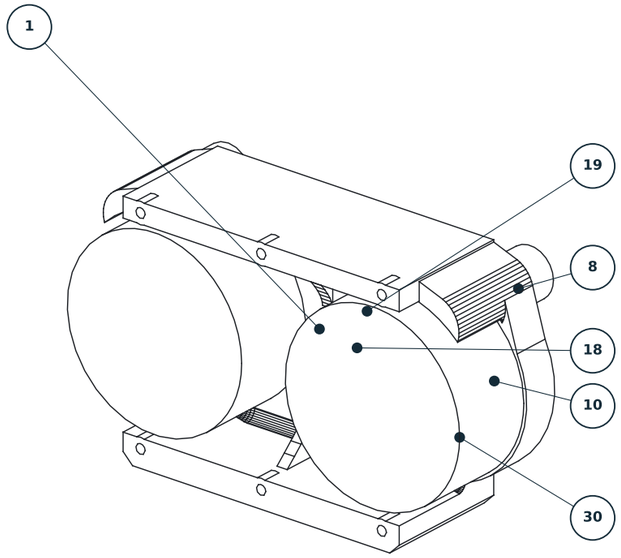
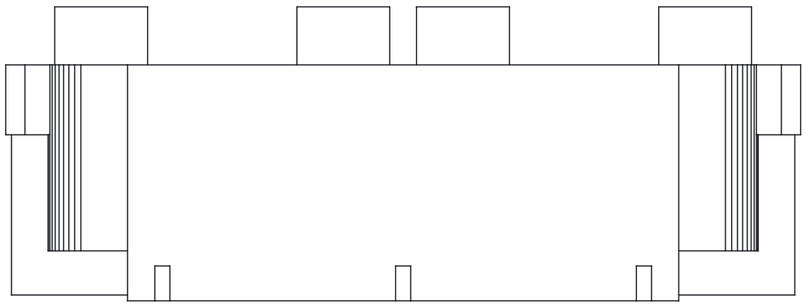


FRONT (LEG SIDE) scale 1:2.1



ISO scale 1:3.3



TOP scale 1:2.1

ITEM	QTY/LEG	PART	SOURCE
1	1	hip frame (PLA)	print: hip_frame.3mf
8	4	stop sleeve (TPU)	print: stop_sleeve.3mf
10	4	Steel Ball Bearing, Steel, Shielded, Trade Numb…	McMaster 6656K236
18	4	Vented Pull-Out Dowel Pin with Flat Shaft, High…	McMaster 98432A386
19	20	Black-Oxide Alloy Steel Socket Head Screw, US O…	McMaster 91290A168
20	4	[not drawn] Dowel Pin, 52100 Alloy Steel, 5 mm …	McMaster 91595A352
24	6	[not drawn] Heat-Set Threaded Inserts for Plast…	McMaster 94459A775
30	2	RobStride RS-04	in hand (arriving 2026-11-02)

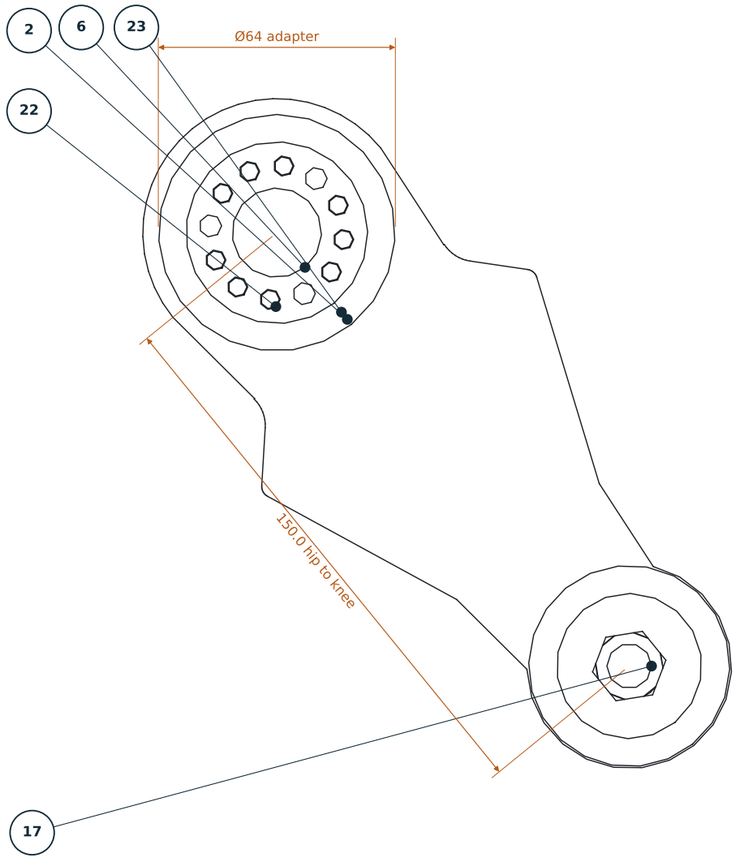
Inserts: six M5 heat-set in the flange ends (item list). Lug screws snug only: the torque is not established (#116). Stop pins bonded with structural epoxy (D74).

GYROPUP **MD-01**

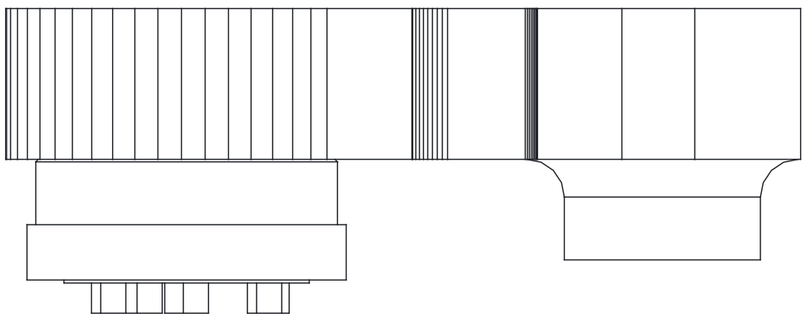
Hip module

the hip frame, the two RS-04s, the support bearings, the stop pins

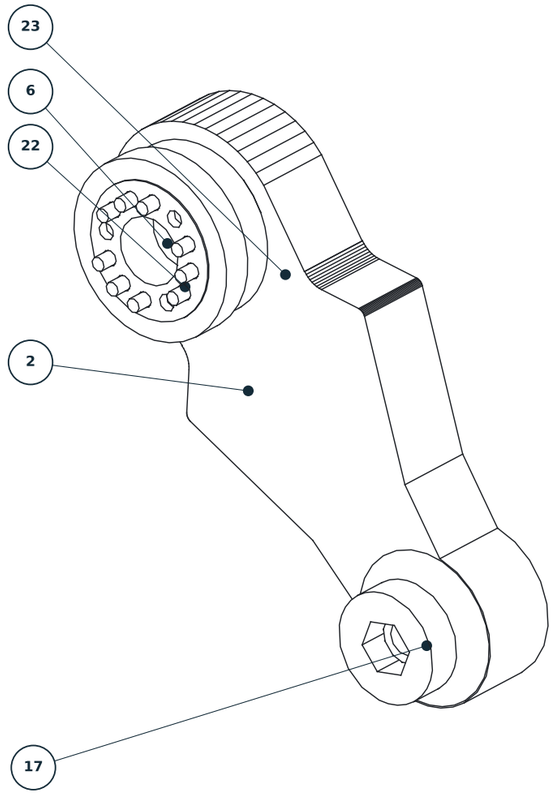
UNITS	mm
SOURCE	8433333
FROM	tools/motor_day.py
STATUS	unqualified: not a release



SIDE scale 1:1.6



TOP scale 1:1.2



ISO scale 1:2.0

ITEM	QTY/LEG	PART	SOURCE
2	2	upper link (PLA)	print: upper_link.3mf
6	2	output adapter pla (PLA)	print: output_adapter.stl
17	4	Metric Press-Fit Drill Bushing, 12 mm ID, 18 mm…	McMaster 96990A403
21	4	[not drawn] Dowel Pin, 52100 Alloy Steel, 5 mm …	McMaster 91595A356
22	18	Black-Oxide Alloy Steel Socket Head Screw, US O…	McMaster 91290A252
23	16	Black-Oxide Alloy Steel Socket Head Screw, M5 x…	McMaster 91290A662

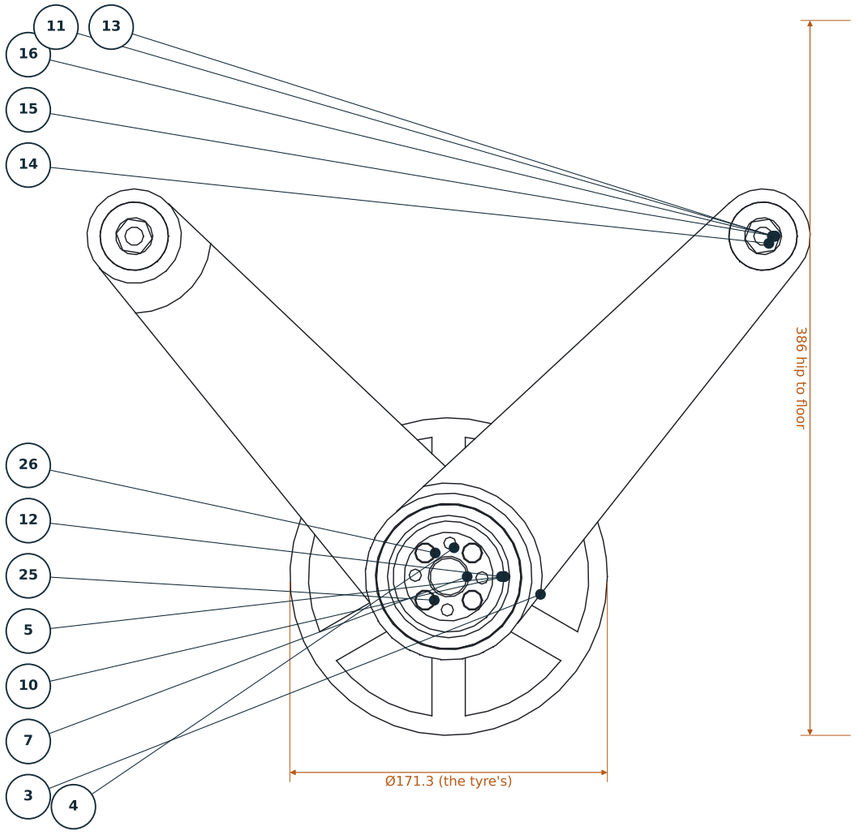
PLA adapter (day one): tap the eight M5 holes by hand; M5 x 40 at 1.0 N m. Replace with the machined 6061 adapter before any run above the PLA cap.

GYROPUP **MD-02**

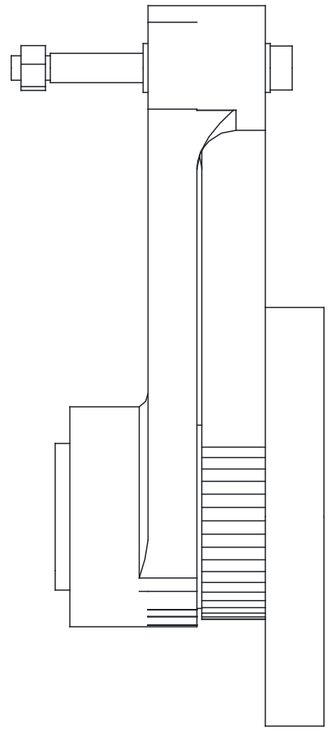
Upper link and adapter

the front side; the rear is its mirror about x = 0

UNITS	mm
SOURCE	8433333
FROM	tools/motor_day.py
STATUS	unqualified: not a release



SIDE scale 1:3.3



FRONT scale 1:2.5

ITEM	QTY/LEG	PART	SOURCE
3	1	front link (PLA)	print: front_link.3mf
4	1	wheel link (PLA)	print: wheel_link.3mf
5	1	apex spacer (PLA)	print: apex_spacer.3mf
7	1	foot (PLA)	print: foot.stl
10	4	Steel Ball Bearing, Steel, Shielded, Trade Numb…	McMaster 6656K236
11	4	Deep groove ball bearing 6301-2RSH (12 x 37 x 1…	McMaster 5972K86
12	1	External Retaining Ring, for 60 mm OD, Black-Ph…	McMaster 98541A174
13	2	Alloy Steel Shoulder Screws, 12 mm Shoulder Dia…	McMaster 92981A411
14	2	High-Strength Steel Nylon-Insert Locknut, Zinc …	McMaster 94645A220
15	4	18-8 Stainless Steel Washer, Undersized, for M1…	McMaster 98689A118
16	2	Multipurpose 6061 Aluminum Round Tube, 0.065" W…	McMaster 9056K68
25	4	Black-Oxide Alloy Steel Socket Head Screw, US O…	McMaster 91290A336
26	4	Washer, undersized (DIN 433), for M6 screw, 6.4…	McMaster (to confirm)
27	4	[not drawn] High-Strength Steel Nylon-Insert Lo…	McMaster 94645A205

Knee nuts 20 N m (steel on steel). Foot: M6 x 40 on DIN 433 washers into nyloc nuts in its pockets, 0.9 N m (PLA under the nuts: re-torque before each session). The foot does not turn: it stands in for the BotWheel at the tyre's radius.

GYROPUP **MD-03**

Lower links, knees, apex and foot

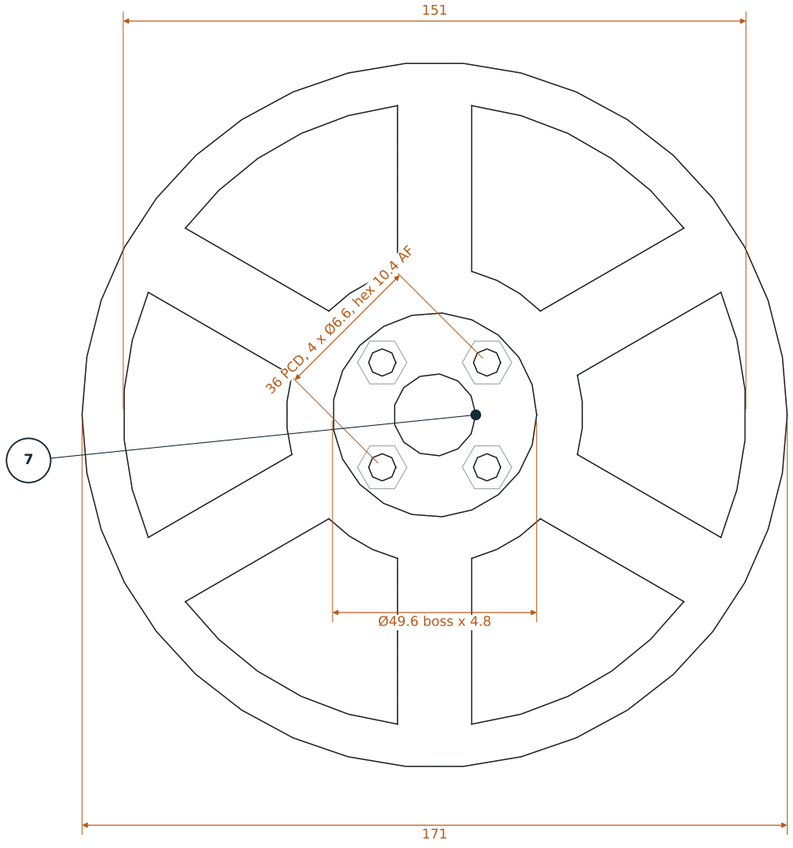
the five-bar's lower half at the 300 mm stance

UNITS	mm
SOURCE	8433333
FROM	tools/motor_day.py
STATUS	unqualified: not a release

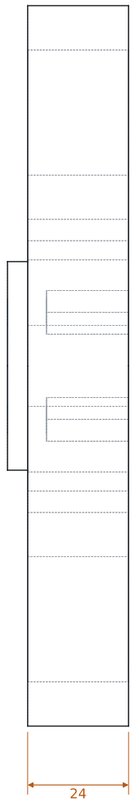
ITEM	QTY/LEG	PART	SOURCE
7	1	Foot, PLA	print: foot.stl / .3mf

Glue a strip of rubber (a bike inner tube) round the rim for grip.

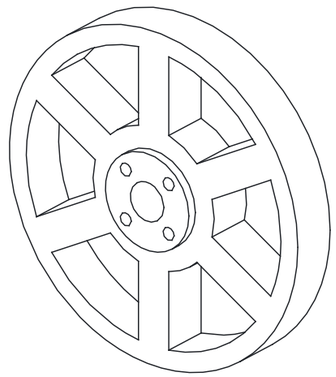
Not in the FEA record: inspect it after every rung.



FACE scale 1:1.5



SIDE scale 1:1.5



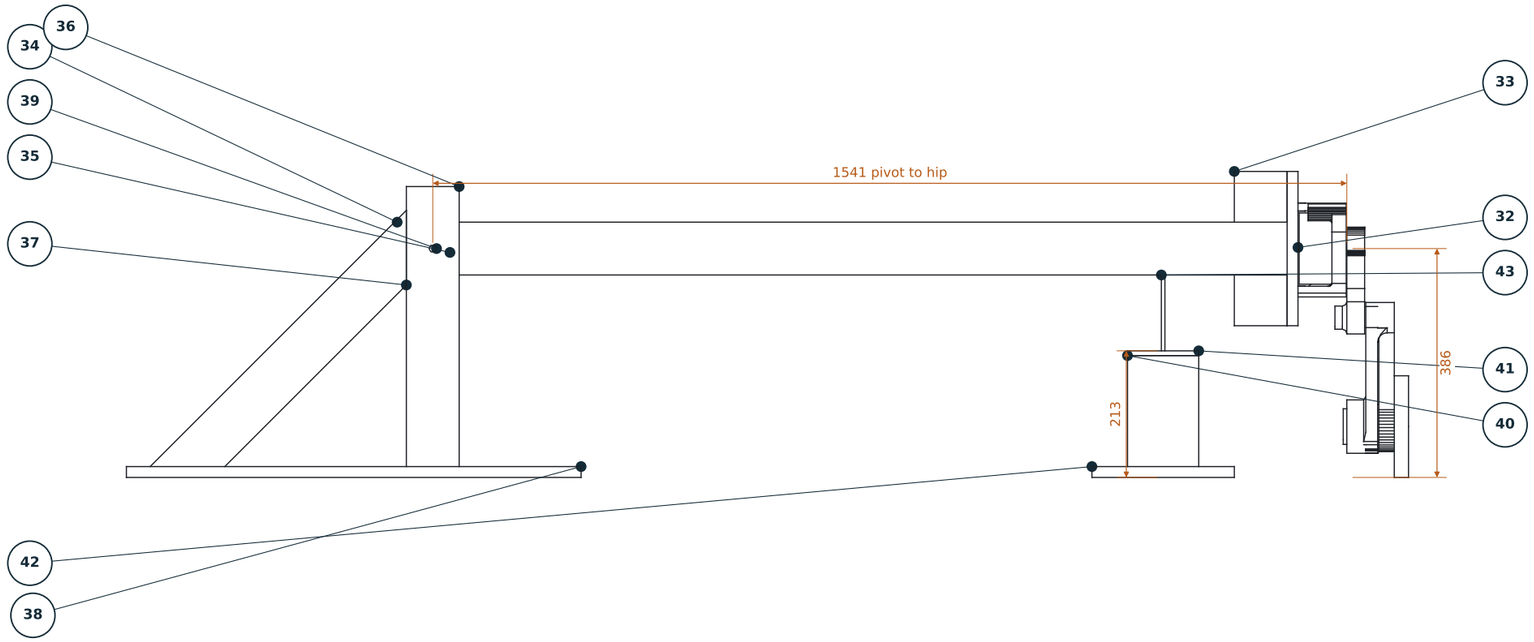
ISO scale 1:2.8

GYROPUP **MD-04**

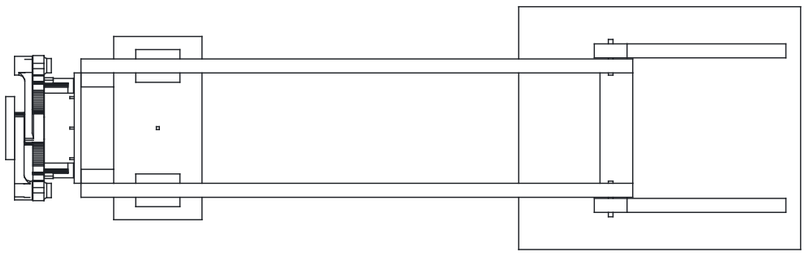
Foot (BotWheel stand-in)

printed PLA, 4 walls, 30 % gyroid; the outer face on the bed

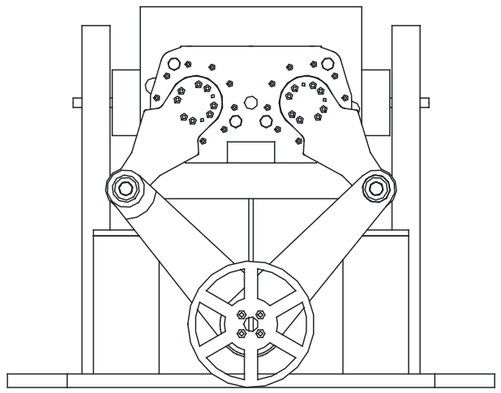
UNITS	mm
SOURCE	8433333
FROM	tools/motor_day.py
STATUS	unqualified: not a release



SIDE scale 1:10.3



TOP scale 1:16.7



LEG END scale 1:8.3

ITEM	QTY/LEG	PART	SOURCE
28	6	[not drawn] Black-Oxide Alloy Steel Socket Head…	McMaster 91290A254
29	6	[not drawn] 18-8 Stainless Steel Oversized Wash…	McMaster 91116A140
32	1	End plate, 300 x 260 plywood (drill through the…	plywood, cut
33	2	Cleat, 2x4 x 260	lumber, cut
34	2	Boom rail, 2x4 x 1500	lumber, cut
35	1	Cross member, 2x4 between the rails' far ends	lumber, cut
36	2	Upright, 2x4 (pivot post)	lumber, cut
37	2	Brace, 2x4 at 45 deg	lumber, cut
38	1	Base, plywood	plywood, cut
39	2	Pivot: 1/2-13 x 5 in carriage bolt, washers, ny…	hardware store
40	2	Floor stop, 2x4 offcuts stacked	lumber, cut
41	2	Rubber pad on the floor stop	rubber mat offcut
42	1	Stop board, plywood	plywood, cut
43	1	Tether: rope and screw eye	hardware store

Floor stop 213 mm tall (rubber pad included): set it so the rails land on it with the leg at 140 mm, before the leg reaches its own stops.

Tether: the hip may rise 150 mm above the stance: at the tether the boom rises 120 mm.

Lumber and plywood densities are estimates: weigh the boom's end on a scale (Phase 3, last step).

GYROPUP **MD-05**

Boom stand

the leg on the end of a pivoting 2x4 boom; the floor stop catches it at a 140 mm leg

UNITS	mm
SOURCE	8433333
FROM	tools/motor_day.py
STATUS	unqualified: not a release